Subject Description Form

Subject Code	**SFT307FI**
Subject Title	**Coloration and Finishing of Fashion & Textiles**
Credit Value	**3**
Level	**3**
Pre-requisite/ Co-requisite/ Exclusion	Exclusion: ITC2203T Foundations of Textiles II
Objectives	The subject provides the basic knowledge and understanding of: (a) Concept and technology of coloration and finishing of different fashion and textile materials (b) Sustainability developments of coloration and finishing technology in the fashion and textile industry (c) Quality control issues related to coloration and finishing technology in the fashion and textile industry
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) Explain the methods and techniques used in the coloration and finishing of different fashion and textile materials; (b) Select suitable substances and explain their functions in coloration and finishing of different fashion and textile materials; (c) Select and describe the suitable machines and technological routes for particular coloration and finishing processes of fashion and textile materials; (d) Evaluate the performance of fashion and textile materials at the different coloration and finishing stages; (e) Apply fundamental principles and instruments for measuring and controlling the colour of fashion and textile materials (f) Describe the methods and standards to communicate colour information (g) Apply their knowledge of the principles and processing involved in coloration and finishing on problem solving of textile materials, and sustainability aspects of the processes.

Subject Synopsis/ Indicative Syllabus	**(I) Preparation of Textile Materials** − Preparation (pretreatment) processes for common textile fibres − Effects of preparation (pretreatment) on the properties of textile materials − Evaluation of quality of materials after preparation (pretreatment) − Ethics and sustainability aspects of preparation (pretreatment) processes **(II) Coloration of Textile Materials** − Introduction of dyeing principles − Applications of different dye classes on common textile fibres − Dyeing operations and related machineries − Auxiliaries for dyeing − Dyebath formulations and design of dyeing processes − Different printing process / application methods − Printing pastes and auxiliaries for printing − Evaluation of performance of colored fashion and textile materials and related problems − Economic aspect of coloration of textile materials − Ethics and sustainability aspects of coloration processes **(III) Finishing of Textile Materials** − Different finishing processes and applications on fashion and textile materials − Working principles and chemicals employed in various finishing processes that impart aesthetic and functional effects − Evaluation of performance of finished fashion and textile materials and related problems − Economic aspect of finishing of textile materials Ethics and sustainability aspects of finishing processes **(IV) Colour Control and Evaluation of fashion and textile materials** − Basic concept of light, object and vision − Quality control issue related to light sources − Methods of colour measurement and communication − Colour quality evaluation

Teaching/Learning Methodology

Lectures will be designed to introduce basic knowledge and understanding of various preparation, coloration, finishing and relatedquality evaluation processes. The principles and practices will be presented to students in a sequential manner. Guidance on further readings and applications will be provided.

Laboratory experiments will be designed to provide students with the opportunities for hands-on practice to help deepen their understanding of the concepts.

Tutorials will be provided to refresh the students' learning and to provide a medium for student to discuss problem-based exercises.

Different assessment methods to consolidate and evaluate the students' learning performance and to improve their problem-solving skills will be employed. They include case analysis on technical problems, written reports, tests, self-study assignment and examination.

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						
		a	b	c	d	e	f	g
Continuous Assessment	**50%**	✓	✓	✓	✓	✓	✓	✓
1. Test	*25%*	✓	✓	✓	✓	✓	✓	✓
2. Report	*25%*	✓	✓	✓	✓	✓	✓	✓
Examination	**50%**	✓	✓	✓	✓	✓	✓	✓
Total	100%							

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Continuous assessment will be used for consolidating and evaluating the students' learning performance. It will help to enhance students' problem-solving skills through case analysis on technical problems, mini-projects of trouble-shooting nature, written reports and tests.

Final examination will evaluate the students' understanding of the basic knowledge of various preparation, coloration, finishing and related quality evaluation processes.

Student Study Effort Expected	Class contact:	
	• Lecture	22 Hrs.
	• Tutorial	4 Hrs.
	• Laboratory	12 Hrs.
	Other student study effort:	
	• Self-study assignment	31 Hrs.
	• Case study	36 Hrs.
	Total student study effort	105 Hrs.

Reading List and References	**Books**
	Gardetti, M.A. and Torres, A.L. (2013). *Sustainability in Fashion and Textiles: Values, Design, Production and Consumption*. Sheffield England: Greenleaf Pub. Hong Kong Cotton Spinners Association (2007), *Textile Handbook: 2007*. 2nd Ed., Hong Kong Cotton Spinners Association, Hong Kong. Ingamells, W. (1993), *Colour for Textiles: A User's Handbook*. The Society of Dyers and Colourists. Roshan, P. (2016), *Denim: Manufacture, Finishing and Applications*. Woodhead Publishing. Heywood, D. (2003) *Textile Finishing*, Society of Dyers and Colourists, Bradford. Clark, M. (2011), *Handbook of Textile and Industrial Dyeing: Volume 1: Principles, Processes and Types of Dyes*. Woodhead Publishing. Clark, M. (2011), *Handbook of Textile and Industrial Dyeing: Volume 2: Applications of Dyes*. Woodhead Publishing. Cie, C. (2015), *Ink Jet Textile Printing*. Woodhead Publishing. Xin, J.H. (2006), *Total Colour Management in Textiles*. Woodhead Publishing

	Blackburn, R. (2015), *Sustainable Apparel Production, Processing and Recycling.* Elsevier. Chakraborty, J.N. (2014), *Fundamentals and Practices in Colouration of Textiles*. Woodhead Publishing India Pvt. Ltd. Gulrajani, M.L. (2010), *Colour Measurement: Principles, Advances and Industrial Applications*. Woodhead Publishing. Miles, L.W.C. (2003), *Textile Printing (2nd ed)*. Society of Dyers and Colourists. **Web-based Materials (registration for free access is required)** Cottonworks (https://www.cottonworks.com/en/) The Woolmark Learning Centre (https://www.woolmarklearningcentre.com/)